Module I: Testing Methodology

#	Question	Marks	Month & Year
1	Discuss verification and validation activities	10	Nov 2019
2	Differentiate between Effective Software Testing and Exhaustive Software Testing	10	Dec 2022
3	Define each software testing terminology: i) Failure ii) Defect iii) Error iv) Testware v) Test oracle	5 (Q1a)	Dec 2022

Module II: Testing Techniques

#	Question	Marks	Month & Year
1	Program reads three numbers A, B, C within range [1,100], prints the minimum — design test cases using BVC and Robust testing	10	Dec 2022
2	Program reads three numbers A, B, C with range [1,50], prints the largest — design test cases using equivalence class testing	10	Nov 2019
3	Explain State Table Based testing with example	5 (Q1d)	May 2023
4	Explain Logic Coverage Criteria for Statement Coverage	5 (Q1c)	May 2023
5	How to test simple loop in a program?	5 (Q1b)	May 2023
6	Explain Bottom-up Integration Testing with one example	5 (Q1a)	May 2023
7	Unit testing and Integration testing (short note)	20 (Q6B)	Nov 2019
8	What is Mutation Testing? Differentiate between primary and secondary mutants	5 (Q1b)	Dec 2022
9	What is Mutation Testing? Explain primary mutant with an example	10	May 2023
10	What is Mutation Testing? Explain Mutation Testing Process	10	Nov 2019
11	C code (prime number program) — draw DD graph, calculate cyclomatic complexity, list independent paths, design test cases	10	Dec 2022
12	Factorial program — draw DD graph, calculate individual cyclomatic complexity for main() and fact(), then whole program	10	May 2023
13	Factorial program — draw DD graph, calculate individual cyclomatic complexity for main() and fact(), then whole program	10	Nov 2019
14	What are graph metrics? Explain with example how to calculate cyclomatic complexity using graph metrics	10	May 2023
15	Comment on regression testing process	10	Nov 2019

Module III: Managing the Test Process

#	Question	Marks	Month & Year
1	Explain structure of testing Group	5 (Q1d)	Dec 2022

#	Question	Marks	Month & Year
2	What are Key elements of Test Management? Explain the structure of testing group	10	May 2023
3	What are Key elements of Test Management? Explain the structure of testing group	10 (Q1B)	Nov 2019
4	What is the need of software measurement? Discuss the various types of software metrics	10	Dec 2022
5	Explain the process of test suite prioritization with its types in detail with example	10	May 2023
6	What is Test Plan? Explain different components of Test plan document	10	Nov 2019
7	What are the components of a test plan? Illustrate test plan hierarchy with a neat diagram	10	Dec 2022
8	Write short note on Test point analysis	20 (Q6A)	Nov 2019
9	Project with distribution of data (SDLC phase, no. of defects, defect age) — calculate defect spoilage	10	Dec 2022

Module IV: Test Automation

#	Question	Marks	Month & Year
1	What criteria will you consider for selection of test tools for automation testing	5 (Q1c)	Dec 2022
2	What is the need of automation testing activities? Differentiate between static and dynamic tools	10	Dec 2022
3	Discuss the features and use of Bugzilla Testing Tool	5 (Q1e)	May 2023
4	Why is automation important in testing? What are the guidelines for selecting testing tools?	10	May 2023
5	Explain the need of automation in testing? Differentiate between manual testing and Automated Testing	10 (Q1A)	Nov 2019
6	Explain a bug life cycle with a neat diagram in detail. List down the states of a bug	10	Dec 2022
7	Classify different types of bugs based on the Software development lifecycle with example	10	May 2023
8	Classify different types of bugs based on Software development lifecycle	10	Nov 2019
9	Bug Lifecycle (short note)	20 (Q6C)	Nov 2019

Module V: Testing for Specialized Environment

#	Question	Marks	Month & Year
1	Explain Agile Testing Life Cycle and its challenges	10	Dec 2022
2	What is Agile testing? Explain challenges in Agile Testing	10	May 2023
3	Explain challenges in Agile Testing	10	Nov 2019
4	What is performance testing? Why is performance testing important?	10	May 2023

Module VI: Quality Management

#	Question	Marks	Month & Year
1	Discuss Six Sigma	5 (Q1e)	Dec 2022

#	Question	Marks	Month & Year
2	Explain McCall's Quality factors and Criteria	10	Dec 2022
3	Explain ISO 9000:2000 standard	10	May 2023
4	McCall's Quality Factors and Criteria (short note)	20 (Q6D)	Nov 2019